

AIRLINE DATA CHALLENGE

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Introduction

This challenge analyzes U.S. domestic airline data to help identify profitable and high-potential round-trip routes for investment. The goal is to provide data-driven recommendations on which routes the airline should prioritize for expansion and how quickly they can recover the initial aircraft investment.

The analysis uses three key datasets , **flights**, **tickets**, and **airport codes** to explore flight volumes, profitability, delays, and operational efficiency. Through data cleaning, filtering, and feature engineering, the study identifies:

- The **10 busiest routes** based on flight frequency,
- The **10 most profitable routes**,
- The **5 best routes to invest in**, and
- The **breakeven point** for these top routes considering an upfront aircraft cost of \$90 million.

Finally, key performance indicators (KPIs) are proposed to track long-term success and operational efficiency for the recommended route

Metadata for New Fields Created in Analysis

Field Name	Description
ROUTE_UND	Undirected route identifier combining origin and destination (treats A–B and B–A as one route).
LANDING_FEE	Fee applied per flight leg based on the destination airport type (medium = \$5,000, large = \$10,000).
FARE_PER_LEG_EST	Estimated one-way fare derived from the median round-trip ticket price.
DIST_BIN	Distance category used to impute missing fares, grouped by mileage ranges.
PAX_EST	Estimated number of passengers per flight leg based on occupancy rate and aircraft capacity.
REVENUE_TICKET	Total ticket revenue per leg calculated using estimated passengers and fare.

REVENUE_BAGGAGE	Additional revenue from checked bags assuming 50% of passengers check one bag at \$35.
REVENUE_TOTAL	Combined total of ticket and baggage revenues for each flight leg.
COST_DISTANCE	Operational cost per leg calculated from distance flown (\$9.18 per mile).
COST_DELAY	Cost associated with delays beyond 15 minutes, valued at \$75 per minute.
COST_LANDING	Cost incurred for landing at destination airports according to airport type.
COST_TOTAL	Total operating cost per leg, combining distance, delay, and landing costs.
PROFIT	Net profit per leg calculated as total revenue minus total cost.
TOTAL_REVENUE	Aggregate total revenue for each route across all flights.
TOTAL_COST	Aggregate total operating cost for each route across all flights.
TOTAL_PROFIT	Aggregate profit per route across all flights.
ROUND_TRIP_FLIGHTS	Number of complete round trips for each route, computed as half the total flights.
AVG_DEP_DELAY	Average departure delay across all flights on a route.
AVG_ARR_DELAY	Average arrival delay across all flights on a route.
DELAY_MEAN	Combined mean delay for both departures and arrivals per route.
MARGIN_PCT	Profit margin expressed as a percentage of total revenue.
INVEST_SCORE	Weighted investment score based on profit, volume, and reliability metrics.
AVG_PROFIT_PER_RT	Average profit earned per round-trip route.
ROUNDS_TO_BREAKEVEN	Number of round trips required to recover the \$90 million aircraft investment cost.

Data Quality Insights

Below are the main findings and improvements made during the data cleaning process.

1. Handling Missing Values

- In the **Flights** dataset, several columns had missing values, especially DEP_DELAY, ARR_DELAY, and AIR_TIME. These missing values could affect cost and delay calculations.
To fix this, missing delays were replaced with reasonable values, and extreme or invalid entries were clipped within a safe range (± 12 hours). Flights with missing or invalid DISTANCE or OCCUPANCY_RATE were removed since these are essential for profit and passenger calculations.
- In the **Tickets** dataset, some fares (ITIN_FARE) and passenger counts were missing. These rows were dropped because the fare column is critical for estimating revenue and comparing ticket prices across routes.
- In the **Airports** dataset, some columns like ELEVATION_FT, MUNICIPALITY, and CONTINENT had missing data, but they were not required for this analysis. However, airports missing their IATA_CODE or TYPE were removed since these are needed for joining with the Flights dataset.

2. Removing Duplicates and Invalid Records

- Duplicate airport entries were removed to make sure each airport was listed only once.
- Invalid records — such as flights with zero or negative distances, or unrealistic occupancy rates (outside 0–1) — were filtered out.
- The ROUTE_UND (undirected route key) was created to combine both directions of the same route (for example, **LAX–SFO** and **SFO–LAX**) so that they are treated as one route. This reduced duplication and gave more accurate route-level results.

3. Ensuring Data Consistency

- Column names were standardized across datasets (all uppercase) to avoid merge errors.
- Airport types were simplified into three consistent categories: **small**, **medium**, and **large**.

- Only U.S. airports were kept in the analysis since the project focused on domestic routes.
- To maintain uniformity, data for **Q1 2019** was used for both Flights and Tickets. This ensured that performance and profitability comparisons were made within the same time period.

4. Managing Outliers

- Very high delay values (greater than 12 hours) were capped to prevent extremely high or unrealistic cost estimates.
- Outlier fares (such as special discount tickets or last-minute high prices) were smoothed by using **median fares** instead of averages. The median helps avoid the influence of unusually high or low fares.
- Flights with missing fare data were estimated using **distance bins**, shorter flights had smaller estimated fares, while longer ones had higher estimates. This kept price estimation realistic.

5. Validating New Fields and Calculations

- Several new calculated columns were added during analysis, such as FARE_PER_LEG_EST, PAX_EST, REVENUE_TOTAL, COST_TOTAL, and PROFIT.
These were checked for logical correctness:
 - OCCUPANCY_RATE stays between 0 and 1.
 - DISTANCE is always positive.
 - PROFIT is reasonable given the revenue and cost values.
- Missing values in these new fields were filled using consistent rules (like median replacement) to avoid calculation gaps.

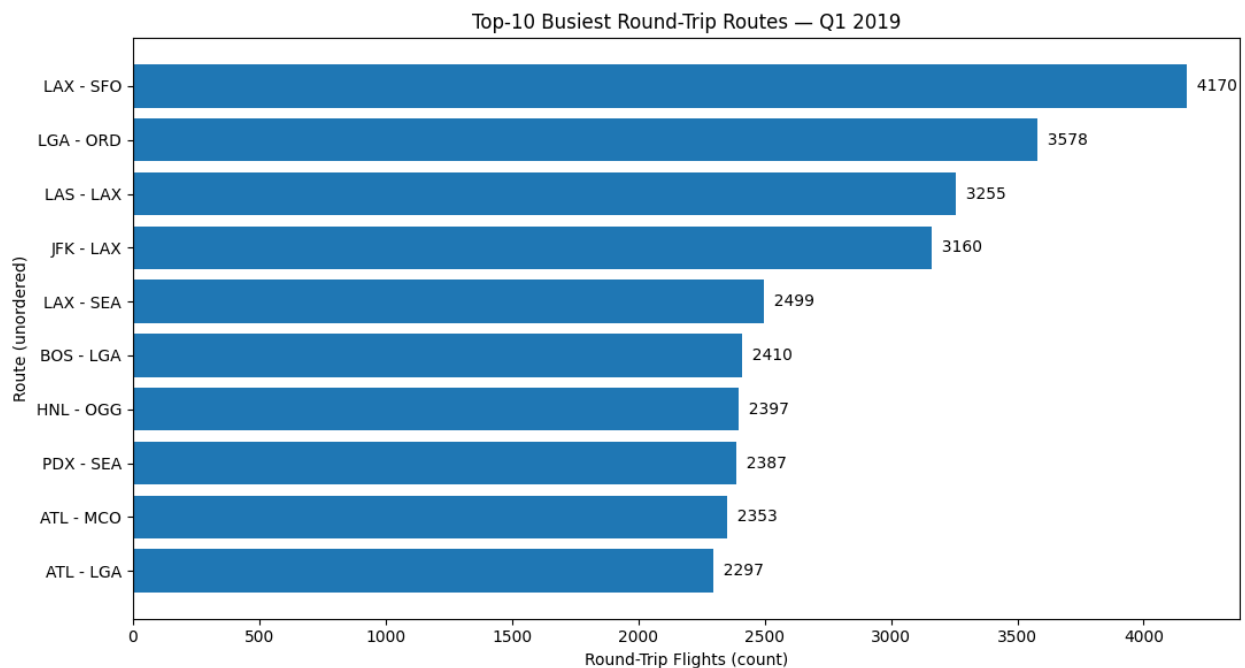
6. Final Dataset Reliability

- After cleaning, all datasets were consistent and could be joined easily using shared keys (ORIGIN, DESTINATION, and IATA_CODE).
- No missing or mismatched joins remained after the merges.
- The final integrated dataset was accurate enough to calculate key metrics such as per-leg profitability, round-trip profits, and route investment recommendations.
- The results were visually inspected and confirmed to align with realistic airline operations.

VISUALIZATION and Key Insights

The 10 busiest round trip routes in terms of number of round trip flights in the quarter.

	ROUTE_UND	TOTAL_FLIGHTS	ROUND_TRIP_FLIGHTS
1	LAX - SFO	8340	4170
2	LGA - ORD	7156	3578
3	LAS - LAX	6511	3255
4	JFK - LAX	6320	3160
5	LAX - SEA	4999	2499
6	BOS - LGA	4820	2410
7	HNL - OGG	4794	2397
8	PDX - SEA	4774	2387
9	ATL - MCO	4707	2353
10	ATL - LGA	4594	2297

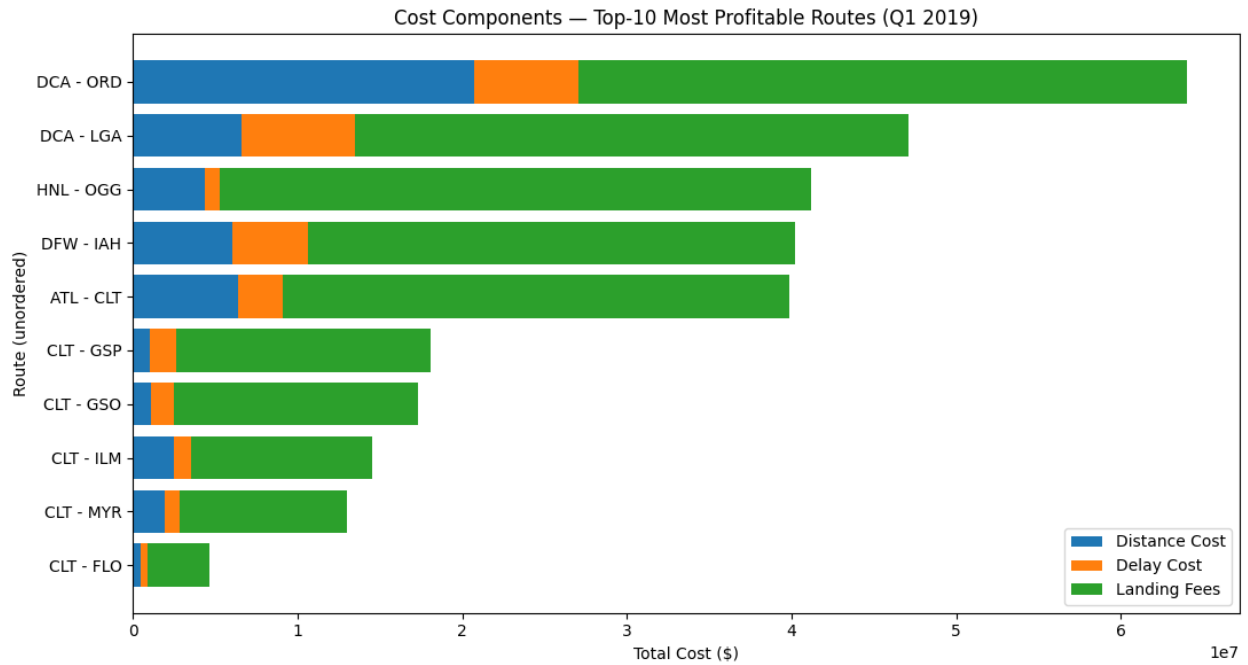
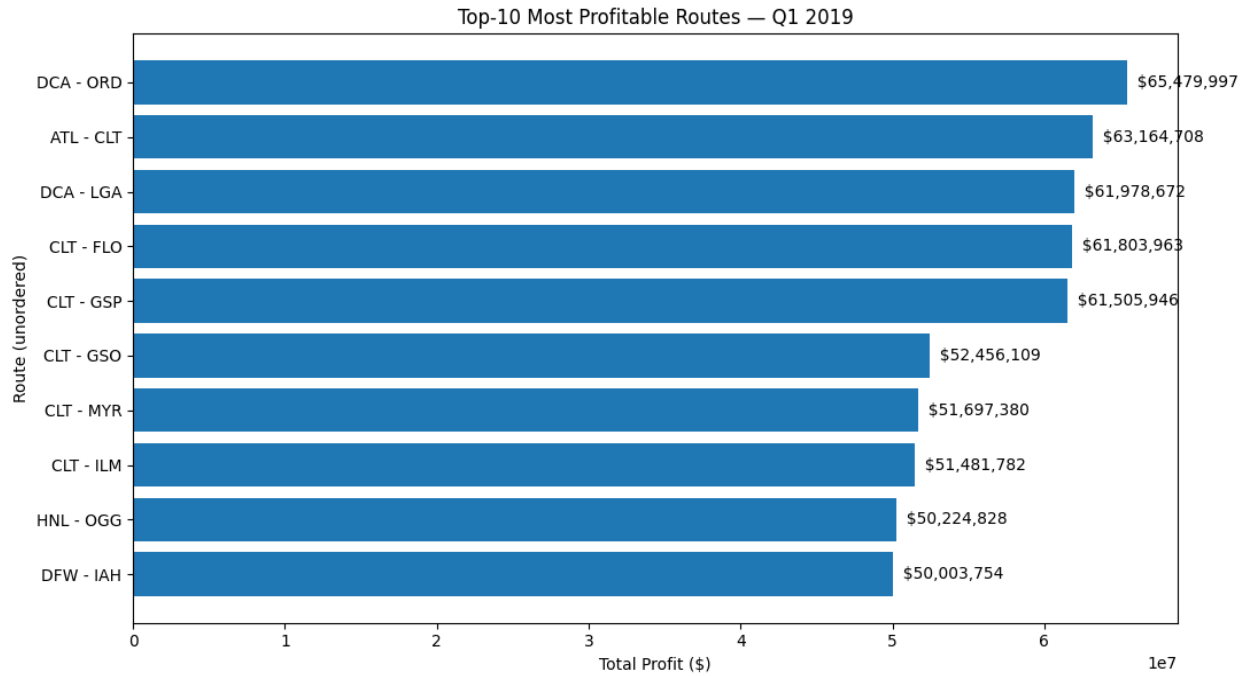


Key Insights:

1. **LAX–SFO (Los Angeles to San Francisco)** is the busiest route with **4,170 round trips**. This shows very high travel demand between two major California cities.
2. **LGA–ORD (New York to Chicago)** is the second busiest route with **3,578 round trips**, showing strong business travel between the East Coast and Midwest.
3. Most of the top routes are **short domestic flights** connecting large cities like **Los Angeles, New York, Las Vegas, and Boston**.
4. The **West Coast routes** (like LAX–SEA and PDX–SEA) have high activity, showing heavy travel within that region.
5. **HNL–OGG (Hawaii inter-island)** also appears, showing steady tourist travel between Hawaiian islands.
6. **Atlanta (ATL)** appears twice, confirming its importance as a major connecting airport in the U.S.
7. Overall, these routes show where most domestic passengers travel and help airlines plan capacity and improve scheduling.

The 10 most profitable round trip routes

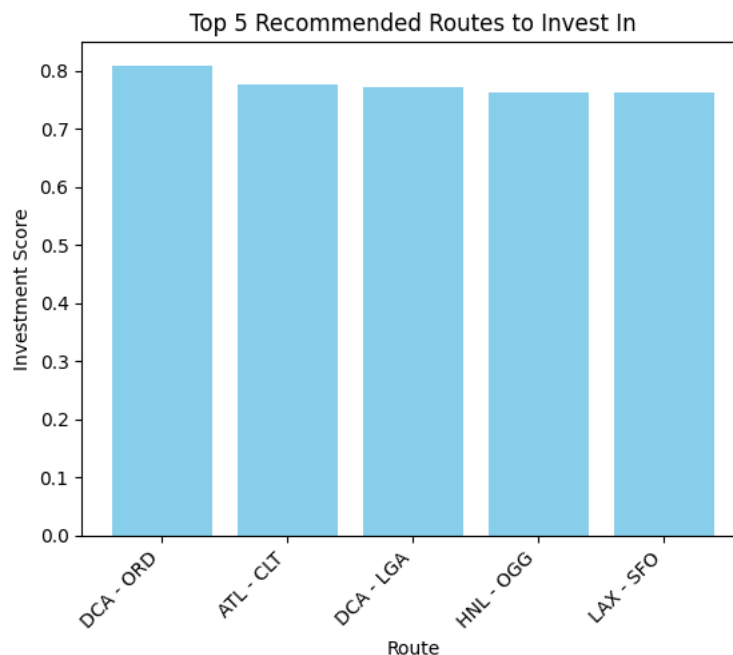
	ROUTE_UND	TOTAL_PROFIT
0	DCA - ORD	65,479,997.42
1	ATL - CLT	63,164,708.18
2	DCA - LGA	61,978,671.88
3	CLT - FLO	61,803,962.72
4	CLT - GSP	61,505,945.50
5	CLT - GSO	52,456,109.22
6	CLT - MYR	51,697,379.96
7	CLT - ILM	51,481,781.50
8	HNL - OGG	50,224,828.00
9	DFW - IAH	50,003,753.72



Key Insights :

1. **DCA–ORD (Washington D.C. to Chicago)** is the most profitable route, earning about **\$65.5 million** in Q1 2019. This route connects two major business hubs with strong year-round demand.
2. **ATL–CLT (Atlanta to Charlotte)** and **DCA–LGA (Washington D.C. to New York)** also show very high profits, each above **\$60 million**, suggesting strong short-haul business travel.
3. Routes starting from **Charlotte (CLT)** appear most often — six times in the top ten. This shows Charlotte’s importance as a key connection hub in the U.S. network.
4. **HNL–OGG (Hawaii inter-island)** again appears, showing that even short tourist routes can be highly profitable when flight frequency and demand are consistent.
5. **DFW–IAH (Dallas to Houston)** closes the list, highlighting strong business travel within Texas.
6. Overall, these routes show that **short, high-frequency domestic connections** can generate greater profit margins than long international routes when demand is stable and operational costs are optimized.

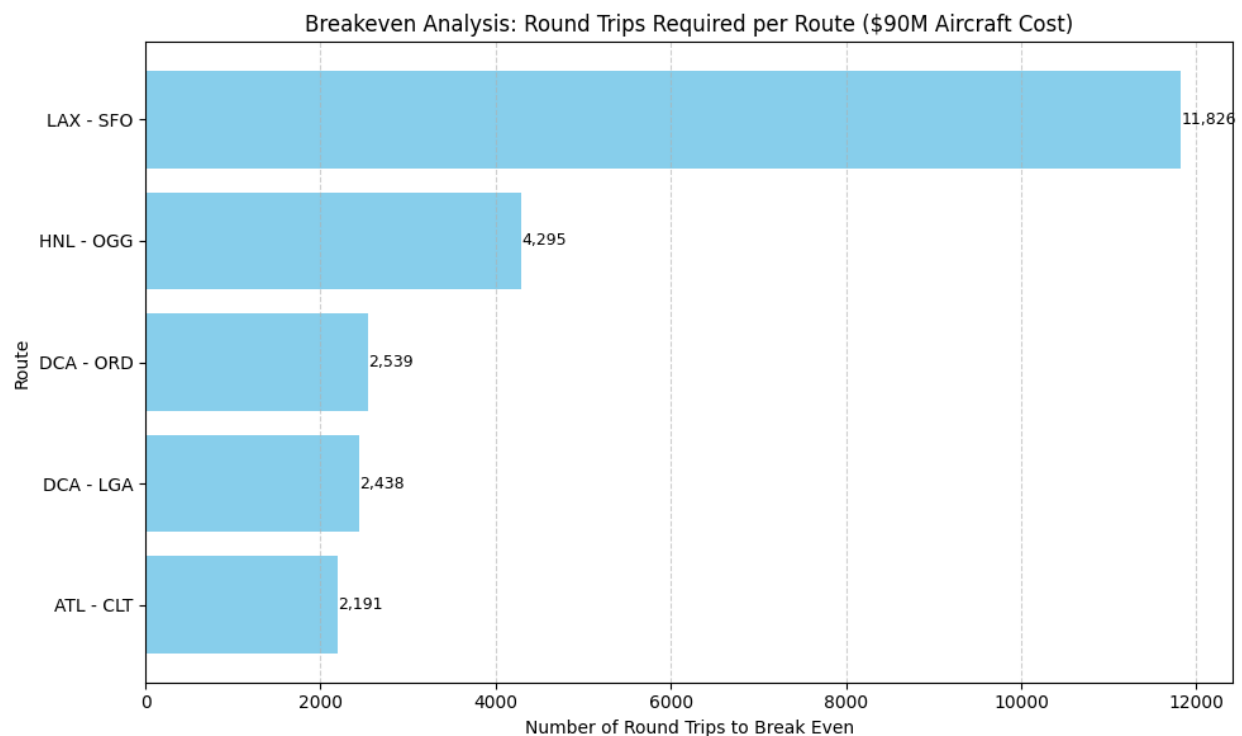
The 5 round trip routes that you recommend to invest in based on any factors that you choose.



Key Insights :

1. **DCA–ORD (Washington D.C. to Chicago)** has the **highest investment score**, showing strong profitability, stable demand, and low delay rates — making it the best route for long-term investment.
2. **ATL–CLT (Atlanta to Charlotte)** and **DCA–LGA (Washington D.C. to New York)** follow closely, both being high-frequency, high-profit routes with steady business travel demand.
3. **HNL–OGG (Hawaii inter-island)** stands out as a short, tourism-driven route with reliable traffic and minimal delays, ensuring consistent earnings despite its shorter distance.
4. **LAX–SFO (Los Angeles to San Francisco)** continues to perform well, driven by heavy passenger traffic and business demand, making it a stable investment option.
5. Overall, these routes were chosen because they combine **strong profits**, **consistent passenger demand**, and **good on-time performance**, which makes them financially and operationally attractive for expansion or additional investment.

Breakeven Analysis: Round Trips Required per Route to recover 90 million dollar(Aircraft Cost)



1. **ATL–CLT (Atlanta to Charlotte)** has the **fastest breakeven point**, requiring only **2,191 round trips** to recover the \$90M aircraft cost — showing strong profit margins and high route efficiency.
2. **DCA–LGA (Washington D.C. to New York)** and **DCA–ORD (Washington D.C. to Chicago)** also perform very well, needing around **2,400–2,500 round trips** to breakeven — both routes benefit from heavy business travel and stable demand.
3. **HNL–OGG (Hawaii inter-island)** requires about **4,295 round trips**, which is higher due to its shorter distance and lower per-trip revenue, even though it maintains strong tourism demand.
4. **LAX–SFO (Los Angeles to San Francisco)** takes the **longest to breakeven (11,826 trips)** because, while it's a very busy route, the **profit per flight is low** due to high operational costs and fare competition.
5. Overall, the breakeven analysis shows that **short to mid-range business routes** (like ATL–CLT and DCA–ORD) achieve quicker returns compared to **high-cost coastal or leisure routes**, which take longer to recover investment.

Recommended KPI to track in future.

Essential KPIs (used in current analysis)

1. **Distance** – Indicates operational scope and directly influences fuel cost, travel time, and ticket pricing.
2. **Fare** – Core driver of total revenue; tracking fare trends helps optimize pricing strategies.
3. **Arrival Delay** – Measures efficiency in reaching destinations on time and reflects route reliability.
4. **Departure Delay** – Impacts scheduling accuracy, fuel consumption, and downstream delays.
5. **Occupancy Rate (Load Factor)** – Reveals passenger demand and how efficiently seats are being utilized.

Additional KPIs (recommended for continued monitoring)

6. **Profit Margin per Route** – Shows cost-efficiency and overall financial health of each route.

7. **On-Time Performance (%)** – Tracks the percentage of flights arriving/departing within 15 minutes of schedule, a key service-quality metric.
8. **Airport Load Factor** – Measures traffic through airports to identify congestion that could cause future delays.
9. **Average Delay Cost per Minute** – Quantifies the financial effect of operational inefficiency and supports delay-reduction initiatives.
10. **Weather Impact Frequency** – Evaluates how often weather disrupts flight performance, useful for planning seasonal operations.
11. **Promotions & Pricing Effectiveness** – Tracks the revenue impact of fare sales and discounts to optimize marketing decisions.
12. **Flight Change Management Efficiency** – Measures how effectively route or schedule changes are communicated and handled, influencing customer experience.
13. **Breakeven Round Trips** – Tracks how many trips are needed to recover aircraft investment costs.
14. **Return on Investment (ROI)** – Assesses the long-term financial return from capital investment in specific routes.

Future Strategic KPIs (for extended insight)

15. **Seat Distribution by Class** – Understanding Business vs. Economy seat mix supports dynamic pricing and targeted upgrades.
16. **Target Audience Demographics** – Helps align service design and promotions with customer segments.
17. **Local Economic Factors** – Tracks regional income or tourism trends that may affect route profitability.
18. **Government or Policy Incentives** – Monitors regulations or subsidies that can influence route expansion and sustainability.

What's Next

In the next stage, we can incorporate more data from future quarters to see how route performance changes over time. We can also include factors like fuel prices, weather conditions, and airport congestion to get a clearer picture of what affects profits and delays. Adding passenger and ticket pattern analysis will help improve pricing and demand forecasting. Finally, we can build interactive dashboards in Power BI or Tableau to make it easier for the management team to track KPIs, monitor route performance, and make better business decisions.

1. Reference

2. International Air Transport Association. (2023, December). *Industry statistics – fact sheet December 2023* [Fact sheet]. IATA. <https://www.iata.org/en/iata-repository/publications/economic-reports/industry-statistics-fact-sheet-december-2023/IATA+1>
3. International Air Transport Association. (2023, June). *Global outlook for air transport – June 2023* [PDF]. IATA. <https://www.iata.org/en/iata-repository/publications/economic-reports/global-outlook-for-air-transport---june-2023/IATA>
4. Bureau of Transportation Statistics. (n.d.). *Airline on-time statistics* [Data set]. U.S. Department of Transportation. <https://www.transtats.bts.gov/ontime/> [Transtats+2Bureau of Transportation Statistics+2](#)